CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

10.1 Practical Circuits

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

A battery is marked 9.0 V.

What does this mean ?

A each coulomb of charge from the battery supplies 9.0 J of energy to the whole circuit

B the battery supplies 9.0 J to an external circuit for each coulomb of charge

C the potential difference across any component connected to the battery will be 9.0 V

D there will always be 9.0 V across the battery terminals

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A battery being marked 9.0 V means that each coulomb of charge from the battery supplies 9.0 J of energy to the whole circuit (this includes the internal resistance of the battery)

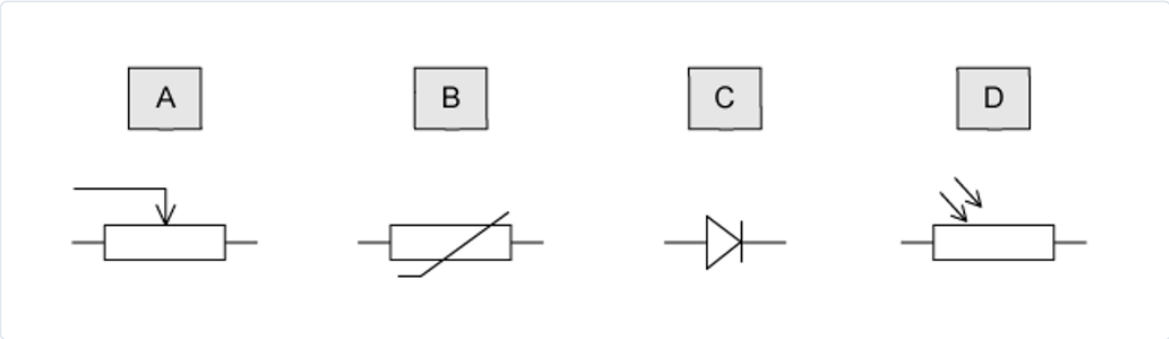
B is incorrect because this would mean that the amount of energy that the battery supplies to an external circuit for each coulomb of charge is not 9.0 V

C is incorrect because if there is no internal resistance in the battery (and the wires have negligible resistance), the sum of p.d. across the components connected to the battery will be 9.0 V

D is incorrect because the e.m.f of a battery does not tell anything about its internal resistance. If the battery has some internal resistance, the p.d. across the battery terminals will not be exactly 9.0 V, but less

Question 2 Easy [1 mark]

Which symbol represents a component whose resistance is designed to change with temperature?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A component whose resistance is designed to change with temperature is called a thermistor
- The symbol for a thermistor is given by option **B**

A is incorrect because this is the symbol for a potentiometer

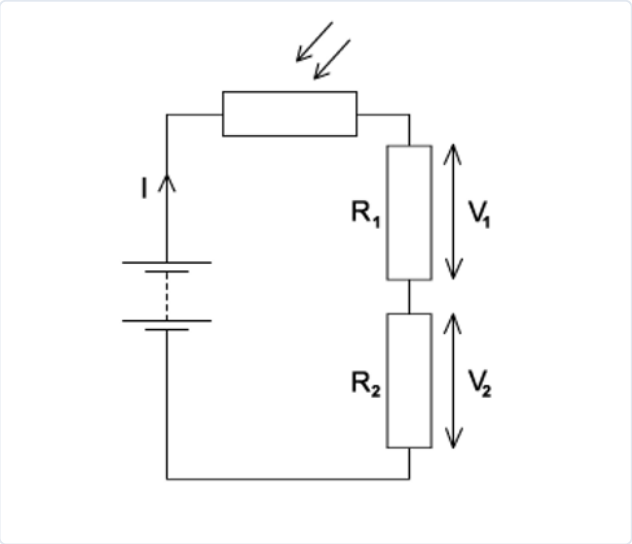
C is incorrect because this is the symbol for a diode

D is incorrect because this is the symbol for an LDR

Question 3 Easy [1 mark]

In the circuit shown, a light-dependent resistor (LDR) is connected to two resistors R_1 and R_2.

The potential difference (p. d.) across R_1 is V_1 and the p .d. across R_2 is V_2. The current in the circuit is I .



Which statement about this circuit is correct ?

A the current I increases when the light intensity decreases

B the LDR is an ohmic conductor

C the p .d. V_2 increases when the light intensity decreases

D the ratio $\frac{V_1}{V_2}$ is independent of light intensity

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The resistance of an LDR decreases as the light intensity increases
- The circuit is a series connection, so the same current flows through all the components
- From Ohm's Law or $V = IR$ or $I = \frac{V}{R}$
- Therefore, as the light intensity decreases, the resistance of LDR increases and so current I decreases
- The ratio of V_1 to V_2 is equal to $\frac{V_1}{V_2} = \frac{IR_1}{IR_2} = \frac{R_1}{R_2}$
- The light intensity affects the current in the circuit, but the ratio does not depend on current

- Therefore, the ratio $\frac{V_1}{V_2}$ is independent of the light intensity

A is incorrect because the current *I* decreases when the light intensity decreases, so light intensity and current are directly proportional

B is incorrect because the LDR is not an ohmic conductor (it does not follow Ohm's Law)

C is incorrect because since current *I* decreases as the light intensity on the LDR decreases, which causes the resistance of LDR to increase, $V_2 = IR_2$ will also decrease

Question 4 Easy [1 mark]

A cell of e. m .f. *E* delivers a charge *Q* to an external circuit.

Which statement is correct?

A the energy dissipation in the external circuit is *EQ*

B the energy dissipation within the cell is *EQ*

C the external resistance is *EQ*

D the total energy dissipation in the cell and the external circuit is *EQ*

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The e.m.f. is defined as the total energy provided by the cell per unit charge
- In equation form, this can be written
  - $E = \frac{W}{Q}$
- So, the total energy dissipation in the cell and the external circuit is equal to
  - $W = EQ$

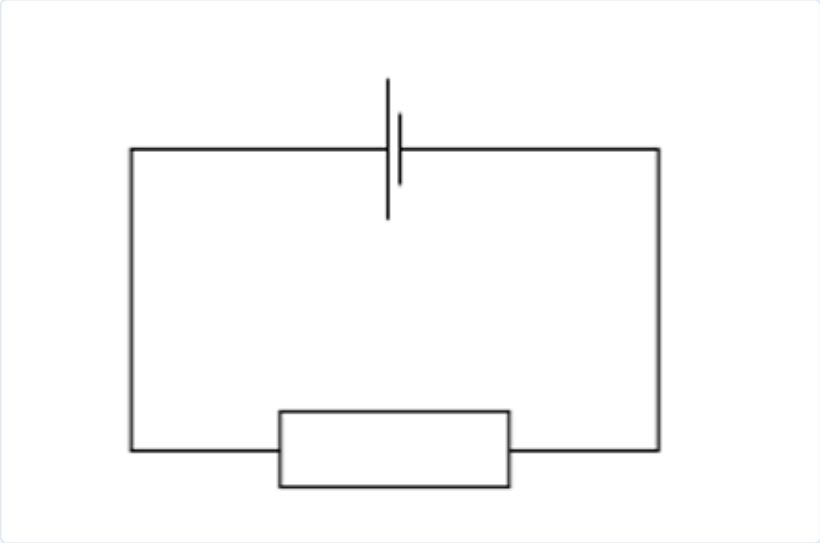
A & B are incorrect because the e .m .f. provides energy across both the cell and the external circuit

C is incorrect because the unit for *EQ* is equal to the joule J, not the Ohm Ω

Question 5 Easy [1 mark]

A cell is connected to a resistor.

At any given moment, the potential difference across the cell is less than its electromotive force.



Which statement explains this ?

A the cell is continually discharging

B the connecting wire has some resistance

C energy is needed to drive charge through the cell

D power is used when there is a current in the resistor

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Most cells tend to have some internal resistance
- The overall e. m. f. provided by the cell is therefore measured as lower than its stated rating
- The e .m .f. of a cell is equal to $E = \frac{W}{Q}$, therefore, energy is needed to drive charge through the cell

A is incorrect because cells discharge over time with continued use, and while this may affect the value of the internal resistance, it is not the reason for the difference between the stated e. m .f. and the measured e. m .f.

B & D are incorrect because the question is in reference to the cell, so the wires and resistor have nothing to do with it

Question 6 Easy [1 mark]

Four statements about potential difference or electromotive force are listed.

1	It involves the transfer of energy to components
2	It involves the transfer of energy into the circuit
3	It is the energy per unit charge to move charge right round a circuit
4	It is the work done per unit charge by the charge moving from one point to another

Which statements apply to potential difference and which apply to electromotive force?

	Potential difference	**Electromotive force**
A	1 and 3	2 and 4
B	1 and 4	2 and 3
C	2 and 3	1 and 4
D	2 and 4	1 and 3

A

B

C

D

Mark Scheme & Explanation:

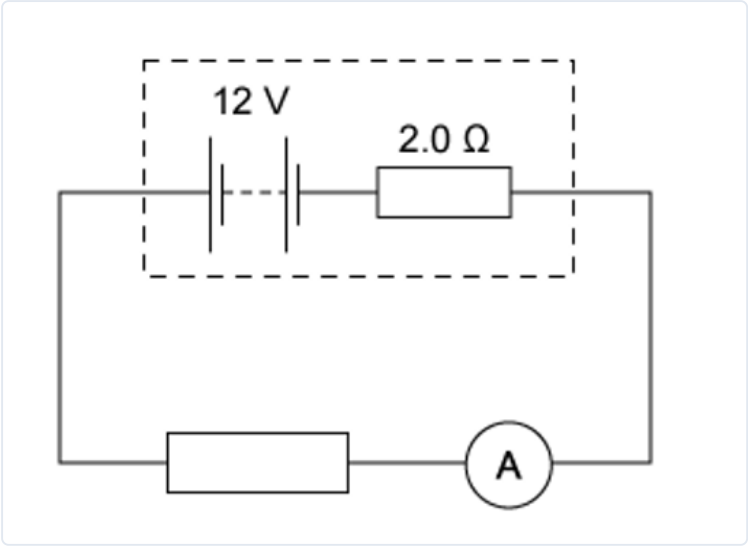
*The correct answer is **B** because:*

- Potential difference (p.d.) is defined as the work done per unit charge by the charge moving from one point to another
- It involves the transfer of energy to components
    - So, statements 1 and 4 refer to p. d.
- Electromotive force (e .m .f.) is defined as the energy per unit charge to move charge right around a circuit
- It involves the transfer of energy into the circuit

- So, statements 2 and 3 refer to e .m .f.

Question 7 Medium [1 mark]

A battery of e .m .f. 12 V and internal resistance 2.0 Ω is connected in series with an ammeter of negligible resistance and an external resistor. External resistors of various different values are used.



Which combination of current and resistor value is **not** correct?

	Current / A	External resistor value / Ω
A	1.0	10.0
B	1.2	8.0
C	1.5	6.0
D	1.8	4.0

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The sum of p. d. across the internal and external resistors is equal to the e. m. f in the circuit
  - $E = I \quad R + r = 12 \quad \text{V}$
  - So, the values must add up to 12 V
- Consider option D

- Internal resistance r = 2.0 Ω
- External resistance R = 4.0 Ω
- Current I = 1.8 A

- $E = I \quad R + r = 1 . 84 + 2 = 10 . 8 \quad \mathrm{V}$
- This is less than the stated e. m. f., so this combination is incorrect

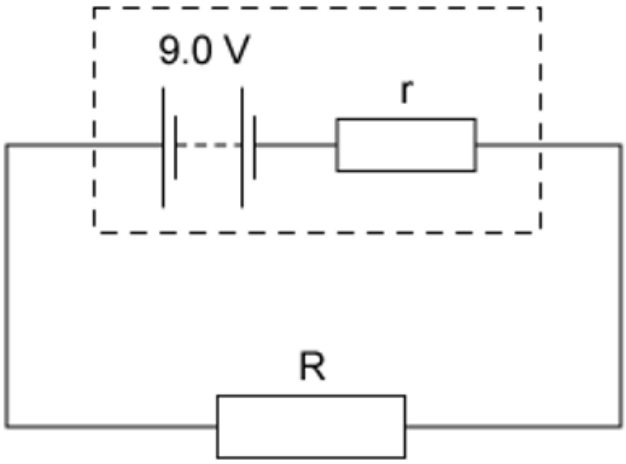
A is incorrect because $E = I \quad R + r = 1 \quad 10 + 2 = 12 \quad \mathrm{V}$

B is incorrect because $E = I \quad R + r = 1 . 28 + 2 = 12 \quad \mathrm{V}$

C is incorrect because $E = I \quad R + r = 1 . 56 + 2 = 12 \quad \mathrm{V}$

Question 8 Medium [1 mark]

A simple circuit is formed by connecting a resistor of resistance *R* between the terminals of a battery of electromotive force (e .m .f.) 9.0 V and constant internal resistance *r* .



A charge of 6.0 C flows through the resistor in a time of 2.0 minutes, causing it to dissipate 48 J of thermal energy.

What is the internal resistance *r* of the battery ?

A 0.17 Ω

B 0.33 Ω

C 20 Ω

D 160 Ω

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Current is equal to $I = \frac{Q}{t}$
- A charge of 6.0 C flows through the resistor in a time of 2.0 minutes
- So, the current through the resistor is $I = \frac{6.0}{2 \times 60} = 0.05 \text{ A}$
- Power dissipated in a resistor is equal to $P = I^2 R$
- Power is also equal to $\frac{energy}{time}$
    - So, energy dissipated $= P \times t = I^2 R \times t = 48 \text{ J}$

- $0.05^2 = R \times 2 \times 60 = 48$
- Therefore, resistance $R = \frac{48}{0.05^2 \times 2 \times 60} = 160 \quad \Omega$

- The internal resistance of the battery and the resistor are in series, hence, the same current flows through them
- So, the sum of p. d. across these two resistors is equal to the e. m .f in the circuit
  - $E = I \times R + r = 9.0 \quad \text{V}$
  - $R + r = \frac{9.0}{0.05} = 180 \quad \Omega$
  - $160 + \quad r = 180$
- Therefore, the internal resistance is equal to r = 180 – 160 = 20 Ω

Question 9 Medium [1 mark]

A battery and a resistor are in series. The e.m.f of the circuit is 15 V and the resistance of the resistor is 8.0 Ω.

Given that the current through the circuit is 1.5 A, what is the value of the lost volts from the battery?

A 3 V

B 4 V

C 9 V

D 12 V

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The known quantities are
  - E.m.f, ε = 15 V
  - Resistance of the resistor, R = 8.0 Ω
  - Current, I = 1.5 A
- The e.m.f of a circuit is equal to
  - $\varepsilon = I\,(R + r)$
- The internal resistance is
  - $\frac{\varepsilon}{I} = R + r$
  - $r = \frac{\varepsilon}{I}$ - R
  - $r = \frac{15}{1.5} - 8.0 = 2 \quad \Omega$
- The lost volts from the battery is
  - $V = I\,r$
  - V = 1.5 × 2 V
- This is option **A**

Question 10 Medium [1 mark]

Which answer option summarises the difference between potential difference and e.m.f?

A Potential difference is the energy transferred into the circuit and e.m.f. is the energy transferred away from the circuit

B e.m.f. is the energy transferred into the circuit and potential difference is the energy transferred out of the circuit

C Potential difference and e.m.f. are chemical energy from the cell converted to electrical energy in the components

D Potential difference is the chemical energy of the cell transferred to the electrical energy of the charges and e.m.f. is the electrical energy of the charges converted to other forms of energy in the components

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Potential difference is the energy per unit charge transferred to the components
- e.m.f. is the energy required per unit charge for the charges to flow against the electrostatic force through the cell
- Therefore, it can be summarised as e.m.f. is the energy transferred into the circuit and potential difference is the energy transferred away from the circuit
    - This is answer **B**

A is incorrect because the descriptions are the wrong way around

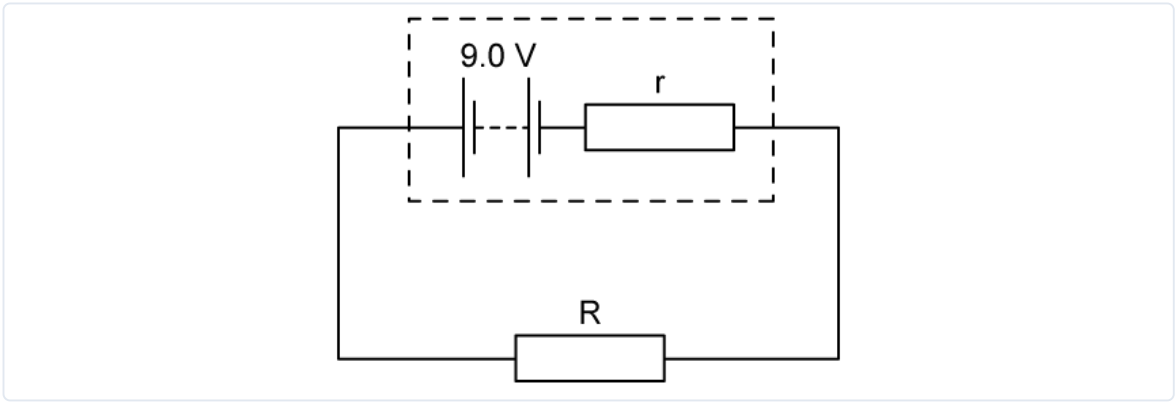
C is incorrect because potential difference and e.m.f. are not the same thing (potential difference and voltage are the same thing)

D is incorrect because the descriptions provided are the wrong way around

- You need to be clear about the difference between potential difference and e.m.f. so take some time to make sure you can explain it

Question 11 Medium [1 mark]

A resistor of resistance R is connected to the terminals of a battery of emf 9.0 V and internal resistance r.



A charge of 240 C flows through the resistor in two minutes. The energy dissipated in the resistor as a result is 1440 J.

What is the internal resistance r of the battery?

A 1.5 Ω

B 2.0 Ω

C 3.0 Ω

D 4.5 Ω

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A charge of 240 C flows through the resistor in a time of 2 minutes = 120 s
- Calculating current:
  - $Q = It \rightarrow I = \frac{Q}{t} = \frac{240}{120} = 2 \text{ A}$
- Power dissipated is $P = I^2R$ and $P = \frac{E}{t}$
  - $\frac{E}{t} = I^2R \rightarrow R = \frac{E}{I^2t}$
  - $R = \frac{1440}{2^2 \times 120} = 3.0 \ \Omega$
- The e.m.f. of the battery is $E = IR + r = 9.0 \text{ V}$
- Therefore, the total resistance in the circuit is $R + r = \frac{E}{I} = \frac{9.0}{2} = 4.5 \ \Omega$

- Since resistance $R = 3.0\ \Omega$, the internal resistance is:
  - $r = 4.5 - 3 = 1.5\ \Omega$

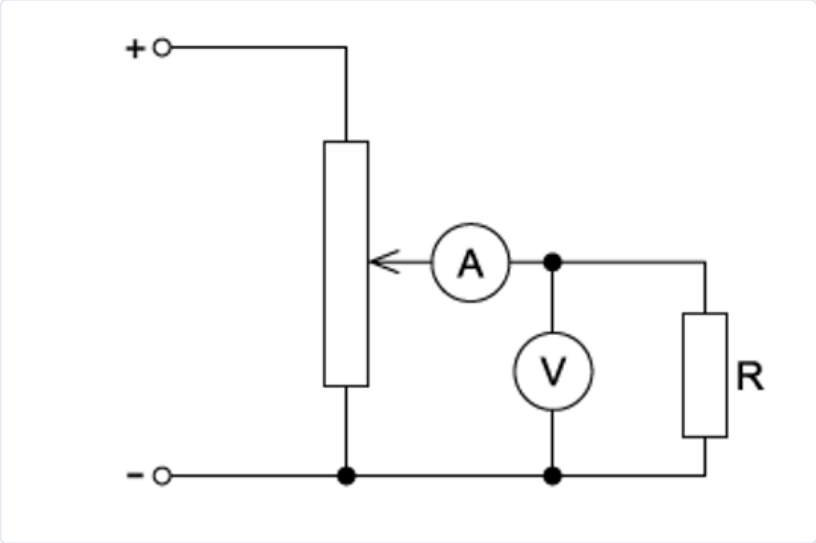
B is incorrect as this is the current in the circuit and not the internal resistance of the e.m.f.

C is incorrect as this is the resistance of the resistor and not the internal resistance of the e.m.f

D is incorrect as this is the total resistance in the circuit and not the internal resistance of the e.m.f.

Question 12 Hard [1 mark]

In the circuit below, a voltmeter of resistance R_V and an ammeter of resistance R_A are used to measure the resistance R of the fixed resistor.



Which condition is necessary for an accurate value to be obtained for R ?

A R is much smaller than R_V

B R is much smaller than R_A

C R is much greater than R_V

D R is much greater than R_A

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- To obtain the resistance of resistor R from the set-up shown, the potential difference and current flowing through the resistor are measured
  - Ohm's law states $V = IR$, so resistance $R = \frac{V}{I}$
- In this circuit the voltmeter is connected in parallel with R (reading the true potential difference across it), while the ammeter sits in the main line and reads the **total** current — the current through R plus the current diverted through the voltmeter
- Ideally, a voltmeter should have infinite resistance so that it draws negligible current; if its resistance is finite, some current bypasses the ammeter reading and the measured value of R is too low

- For the diverted current to be negligible, the voltmeter resistance must be much larger than R, so that $I_V \ll I_R$
- Therefore, in order to obtain an accurate value of R, the necessary condition is that R should be much smaller than R_V

B & **D** are incorrect because R_A does not affect the result. The voltmeter reads the potential difference across R directly, and the ammeter sits in the main supply line, so its resistance R_A only reduces the total current drawn from the supply — it does not change the ratio $\frac{V}{I}$, and so the measured value of R is unaffected by whether R_A is small or large

C is incorrect because an ideal voltmeter has infinite resistance, so R could not be higher than this

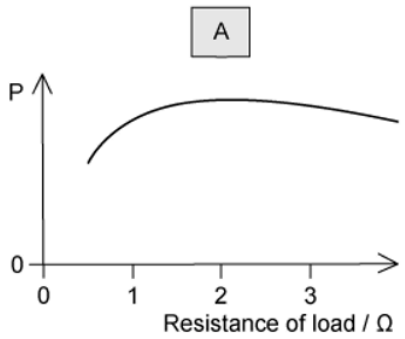
Note that the ammeter is not in series with resistor R alone — it is in series with the parallel combination of the voltmeter and the resistor. So the ammeter measures $I_R + I_V$, not the current through R

Question 13 Hard [1 mark]

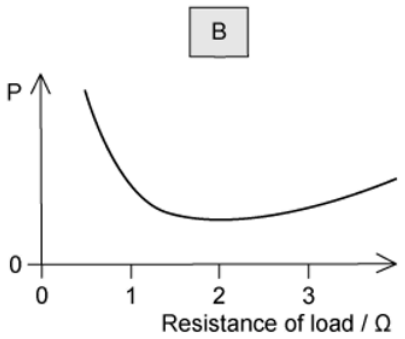
A power supply of electromotive force (e .m .f.) 12 V and internal resistance 2 Ω is connected in series with a load resistor. The value of the load resistor is varied from 0.5 Ω to 4 Ω.

Which graph shows how the power *P* dissipated in the load resistor varies with the resistance of the load resistor ?

A



B



C



D



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The internal resistance of the supply is r = 2 Ω
- The resistance of the load resistor, R varies from 0.5 Ω to 4 Ω

- Power *P* dissipated in the load resistor is equal to I^2R
- The power dissipated is maximum when the load resistance is equal to the internal resistance, $R = r = 2\ \Omega$
- Since 2 Ω lies within the range 0.5 Ω - 4 Ω, the graph peaks at 2 Ω, as illustrated in option **A**

B is incorrect because this graph shows that the power reaches a minimum value when the load resistance equals the internal resistance

C & D are incorrect because at a glance of the equation between power and resistance $P = I^2 R$, you might initially think the relationship is directly proportional. However, we have to consider the internal and external resistors as two resistors in series. Initially, the value of *R* is zero, so the power dissipated across it is also zero. Similarly, if the value of *R* is infinite, the power dissipated is also zero as the current will then also be zero. So, the power dissipated must increase and then decrease as *R* increases from zero. As the value of *R* increases, the current *I* decreases and hence the p. d. across the external resistor increases. The p .d. across the internal resistor *r* decreases and the power dissipated in *r* (given by $I^2 r$) decreases. Therefore, the maximum power dissipated across R occurs when $R = r$

Question 14 Hard [1 mark]

A relay is required to operate 800 m from its power supply. The power supply has negligible internal resistance. The relay requires 16.0 V and a current of 0.60 A to operate.

A cable connects the relay to the power supply, and two of the wires in the cable are used to supply power to the relay.

The resistance of each of these wires is 0.0050 Ω per meter.

What is the minimum output e .m .f. of the power supply ?

A 16.6 V

B 18.4 V

C 20.8 V

D 29.3 V

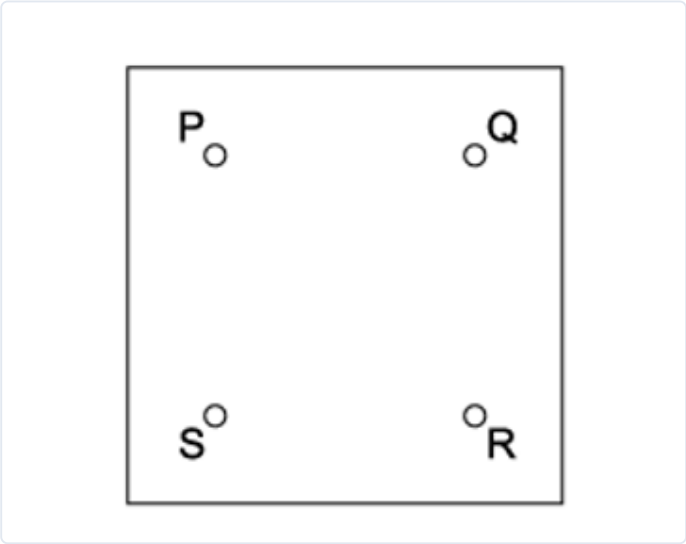
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In the cable, two of the wires are used to supply power to the relay
- So, the circuit is connected as follows:
  - One terminal of power supply — 1st wire — relay — 2nd wire — other terminal of power supply, therefore, it is a series connection
- The resistance of each of these wires is 0.0050 Ω per meter and each wire measures 800 m long
  - So, the total resistance of one wire $= 800 \times 0.0050 = 4 \quad \Omega$
  - Therefore, the total resistance due to both wires = 4 + 4 = 8 Ω
- There will be a p.d. across the wires and since this is a series connection, the same current flows through the whole circuit
  - So, the p.d. across the two wires $= IR = 0.60 \times 8 = 4.8 \quad \text{V}$
- The sum of p.d. across the whole circuit should be equal to the minimum e.m.f
  - Therefore, the minimum output e.m.f. = 16.0 + 4.8 = 20.8 V

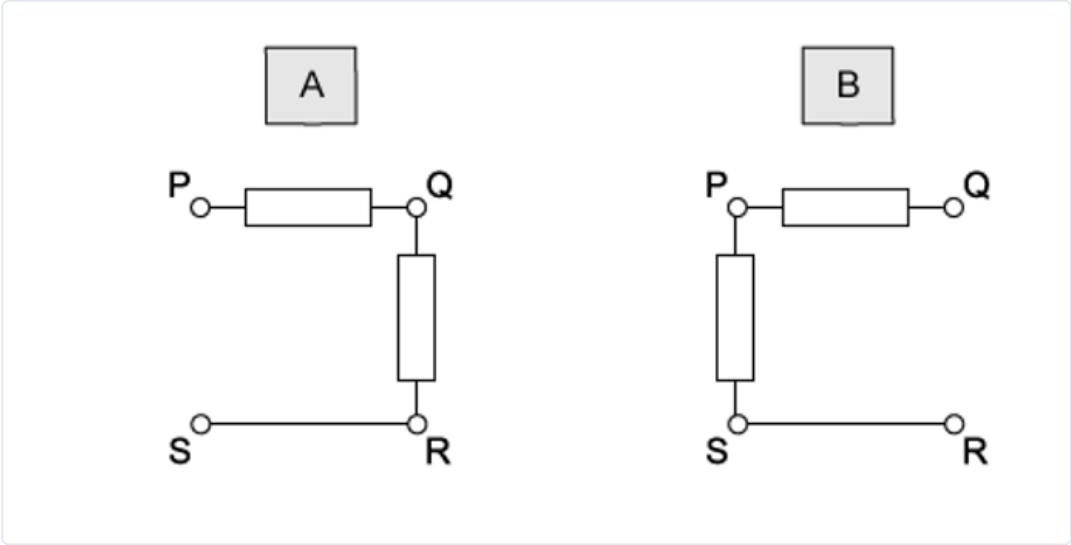
Question 15 Hard [1 mark]

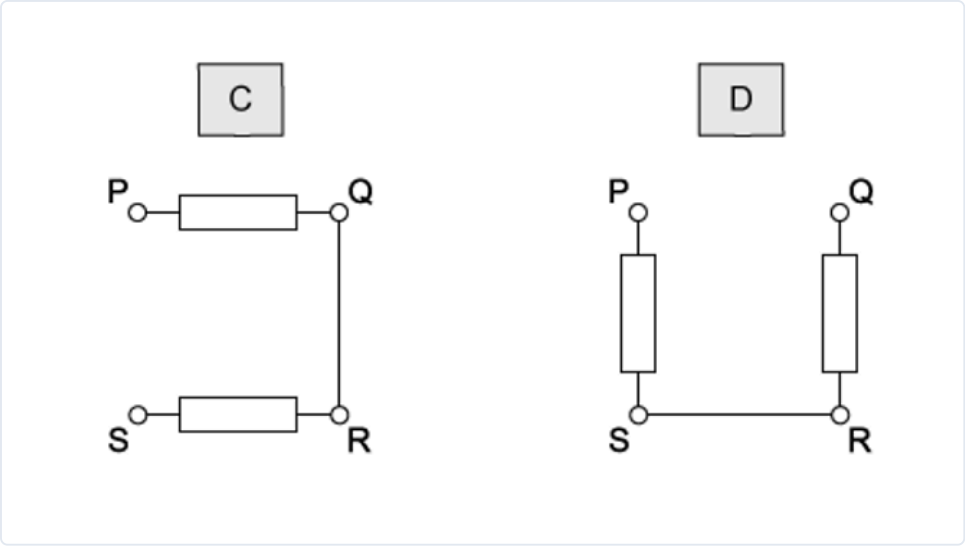
A box with four terminals P, Q, R and S contains two identical resistors.



When a battery of electromotive force (e .m. f.) *E* and negligible internal resistance is connected across PS, a high-resistance voltmeter connected across QR reads $\frac{E}{2}$

Which diagram shows the correct arrangement of the two resistors inside the box ?





A

B

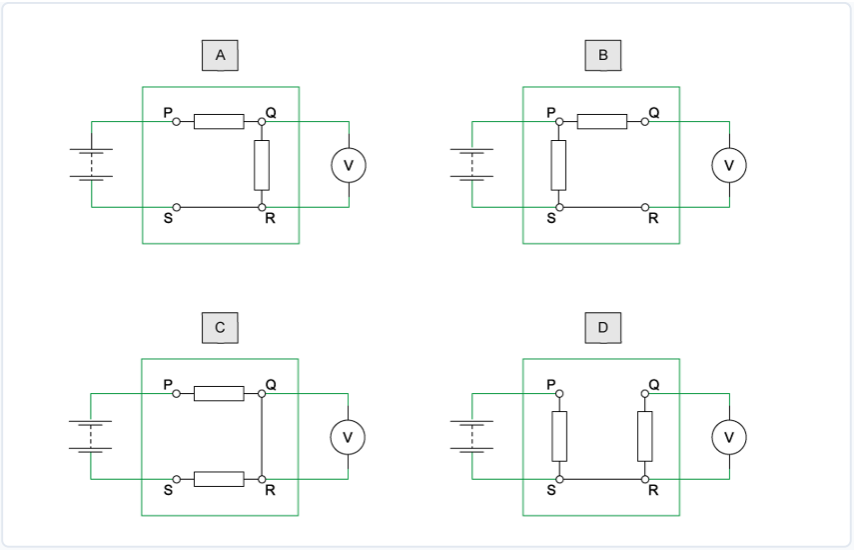
C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The best approach to this question is to draw the battery and voltmeter in each of the four circuits, as shown below
- The reading on the voltmeter across QR is $\frac{E}{2}$
- Ohm's Law states $V = IR$
  - This means that half of the resistance, or one resistor, is across QR
- Consider circuit **A**:
  - The voltmeter is connected across one of the two resistors, and the circuit is complete
  - The reading on the voltmeter would therefore be $\frac{E}{2}$
  - So, option A is correct

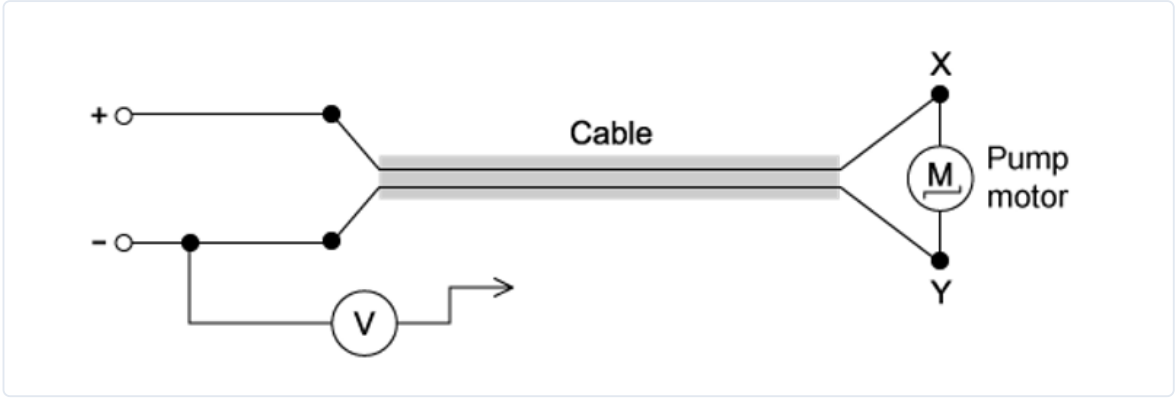


B & **C** are incorrect because to get a reading of $\frac{E}{2}$ there would need to be just one resistor across QR

D is incorrect because this is not a complete circuit, so the reading would be zero

Question 16 Hard [1 mark]

The diagram shows the electric motor for a garden pump connected to a 24 V power supply by an insulated two-core cable.



The motor does not work so, to find the fault, the negative terminal of a voltmeter is connected to the negative terminal of the power supply, and its other end is connected in turn to terminals X and Y at the motor.

Which row represents two readings and a correct conclusion?

	Voltmeter reading when connected to X / V	Voltmeter reading when connected to Y / V	Conclusion
A	24	0	break in positive wire of cable
B	24	12	break in negative wire of cable
C	24	24	break in connection within the motor
D	24	24	break in negative wire of cable

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- From the diagram, X is connected to the positive terminal of the power supply, and Y is connected to the negative terminal of the power supply
- So, if the potential at X is 24 V, which it is for all options, then there must be no break in the positive wire
- Now, consider a break in the negative wire of the cable
- The circuit is not complete, and so, there is no current passing through the circuit
- So, there is no drop in potential across the pump motor (in other words, the potential at Y is the same as the potential at X)
- Therefore, a break in the negative wire of the cable would give a potential of 24 V at Y, as well as 24 V at X

A is incorrect because if there was a break in the positive wire, the voltmeter reading when connected to X or Y would be zero

B is incorrect because if there was a break in the negative wire, the potential at Y would be the same as the potential at X (24 V)

C is incorrect because if there was a break in the connection within the pump motor X would be at 24 V and Y would be at 0 V